Output Diversity and Cross-Model Convergence in Preference-Aligned Language Models

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Abstract. When AI language models are trained to satisfy human preferences, do their outputs become too similar to each other? Even on open-ended questions where no single answer exists, this study finds they do. We call this risk AI-induced monoculture.
We test five preference-aligned models against human responses in three domains: open-domain QA, finance, and creative writing. On 87–98% of prompts, four of these models (one from each of four independent developers) answer more like *each other* than like humans: a convergence of register (shared writing style) and, on the prompts we hand-checked, of content. The pattern is strongest in finance and holds across question types, including the advice-and-opinion questions where diversity matters most and a single correct answer is least appropriate (98.3% convergent).
The sharpest test uses matched data: where a corpus provides several human answers to the same prompt, we ask whether the models cluster more tightly than humans do. On 97.8% of finance prompts (FiQA, Cohen's $d = 2.74$) and 94.3% of writing prompts ($d = 2.05$), they do: the models agree with each other more than humans agree with each other. A manual check confirms the convergence reflects real content similarity on most prompts, though the metric can overstate it in some cases. Comparisons against unaligned base models in two independent model families point to the post-training pipeline, not shared pre-training data, as the driver, though that test runs at the 8B scale rather than on the largest models.
What matters is not average diversity across a corpus but what happens on a single prompt: a model can look varied overall yet give the same answer to every user who asks the same question. We argue this per-prompt convergence should be a standard part of pre-deployment evaluation, which current quality, diversity, and safety checks do not capture.

Keywords: Output diversity · Preference alignment · RLHF · Large language models · AI-induced monoculture · Text homogenisation · Model evaluation.

1 Introduction

Generative AI has become infrastructure. ChatGPT reached 100 million users within two months of release, and language models now help with scientific writing, diagnosis, legal reasoning, and creative work at a scale no earlier technology reached [14]. They also produce a large and growing share of the text people read. These systems are trained to produce the *most plausible* output, the core objective of maximum likelihood estimation (MLE). Preference-based alignment, whose best-known form is reinforcement learning from human feedback (RLHF) [19], then pushes them further toward the outputs a majority of human raters prefer [17].

For tasks with a correct answer, such as drug interactions, legal compliance, or factual retrieval, this convergence is useful. The problem is that AI systems apply the same compression *uniformly*, including in domains where a range of perspectives is the point.

Two of this study's three domains fit that concern directly. The first is open-domain QA, which people use daily to form views about the world. The second is finance, where they ask AI for help with important decisions. These questions rarely have one correct answer. The third domain, creative writing, is included as a contrast: there a narrow range of outputs is a more visible failure, so it tests whether any narrowing is confined to information-seeking use or extends to open-ended generation. If AI answers converge because the models were trained to agree, the range of framings in circulation shrinks, a narrowing invisible to any single user and visible only in aggregate. Ashkinaze et al. [3] showed experimentally that high AI exposure makes ideas more similar *across* people without reducing any individual's creativity. This study measures the condition in the text that could make such an effect possible.

This broader risk is what this study calls **AI-induced monoculture**. That societal end state is not tested here. What is tested is a measurable upstream condition (Figure 3): output convergence, a *necessary but not sufficient condition* for monoculture. Whether aligned models produce less diverse text than humans, and whether they converge on similar answers to the same prompt, is directly measurable. Whether that convergence narrows collective thought is motivated by outside evidence, not established here.

Research questions. This study addresses four research questions:

RQ1 (Diversity Gap): Do aligned models produce less diverse responses than humans on the same prompts, across semantic, lexical, and compression-based dimensions, and does the gap replicate across models from different organisations and architectures?

RQ2 (Domain Generality): Does the gap hold across multiple domains, including creative writing, where there is no single correct answer?

RQ3 (Alignment Attribution): Does the post-training alignment pipeline contribute to the reduction?

RQ4 (Per-Prompt Convergence): On a given prompt, do aligned models produce responses more similar to each other than to the human response on that prompt?

RQ1–RQ4 are tested by comparing five preference-aligned models (ChatGPT, Llama-3.1-8B-Instruct [7], Qwen3-32B-Instruct [21], Gemma-3-27B-it [11], and Apertus-8B-Instruct [2]) against human-written text from HC3 [12] and

Reddit WritingPrompts [9]. For RQ3, each instruct model is compared against its own unaligned base version in two families (Llama-3.1-8B [7] and Mistral-7B [16]). Diversity is measured across three dimensions: semantic distance, lexical diversity, and compression-based redundancy [26, 24].

Summary of findings. The headline result is per-prompt: aligned models answer more like each other than like the human on the large majority of prompts in every domain (Figure 1, Section 5.3). Corpus-level metrics agree in open-domain QA and finance for every model (Figure 6), with the vocabulary and compression gaps partly length-inflated in open-domain QA but not the semantic-diversity gap (Appendix G). Creative writing is mixed. The instruct-versus-base comparison further implicates alignment (Section 5.2). Throughout, corpus-level spread *across* prompts and per-prompt convergence *on the same* prompt are kept separate: a model can look diverse on average yet converge whenever two systems face the same question.

Contributions. Four contributions, ranked by the strength of the supporting evidence. *Primary:* a **per-prompt convergence analysis and a diagnostic built from it** (RQ4). Aligned models answer a prompt more like each other than like the human on 87–98% of prompts across all three domains, the result that maps directly onto what a user actually receives. We package this into a concrete, reusable tool (Figure 2): a cheap pre-deployment check that flags per-prompt sameness hidden behind healthy corpus-level diversity, a failure mode that standard quality, diversity, and safety evaluations miss, and which we recommend reporting in model cards. The metric also gives mitigation work (pluralistic alignment) a concrete quantity to measure against. *Secondary:* **cross-model convergence evidence**: the per-prompt and corpus gaps are cross-model (between independent developers), and independent models share a measurable register (shape analysis, Section 5.5), pointing to a shared attractor rather than coincidence. *Tertiary:* **controlled alignment attribution** in two instruct/base families (Llama-3.1-8B, Mistral-7B): semantic- and compression-diversity reductions replicate in both, while the vocabulary effect is family-specific. *Exploratory:* a structured framing of **AI-induced monoculture** as a testable societal risk, where output convergence is a necessary but not sufficient upstream condition, with the downstream step motivated but not established.

2 Background

Large language models (LLMs) are trained to predict the next token by minimising cross-entropy loss over large text corpora (maximum likelihood estimation, MLE). Under MLE, frequent patterns get high probability and rare ones low, whether or not they are valid: the model is rewarded for matching its training data, not for novelty.

A pre-trained LLM is fluent but not reliably helpful or safe. The post-training alignment pipeline fixes this: supervised fine-tuning (SFT) on example answers, then preference-based optimisation, which may be RLHF [19], DPO, or another variant. All steer the model toward the outputs most raters prefer. Because the five models here were aligned with different algorithms, the umbrella term *preference-based alignment* (or just alignment) is used, with RLHF kept only for models that use it. The algorithms differ but share one shape: a preference signal trained on *majority* human judgments moves the model away from its raw MLE output, a second pressure toward sameness on top of MLE. That shared property is what this study measures the effect of.

Textual diversity is not one thing: answers can use varied words yet mean the same, or the reverse. Three metric families capture different sides. N-gram metrics count surface overlap but miss convergence in meaning; embedding metrics place each text as a vector so similar meanings lie close, matching human diversity judgements better [26]; compression metrics measure repetition directly at low cost [24]. This study uses one per family, semantic diversity (SD), lexical diversity (LD), and compression ratio (CR), each defined in Section 4.

Four terms recur and are kept distinct. The *diversity gap* is corpus-level: how much a source's answers vary across many prompts. *Per-prompt convergence* is within-prompt: how alike two models' answers to the *same* prompt are. *Compression* is one specific redundancy metric (CR), not the phenomenon itself. *AI-induced monoculture* is the broader, untested societal risk that motivates the study, not a measured result.

3 Related Work

Four strands of prior work inform this study: the statistical properties of language model training and decoding, how alignment affects output diversity, the downstream effects of widespread AI use on human expression, and existing approaches to preserving diversity.

3.1 MLE Training and the Tendency Toward Average Outputs

MLE-trained models favour dominant patterns over rare ones: Bender et al. [4] call LLMs "stochastic parrots" that amplify the skew of their training corpora, before any alignment. Holtzman et al. [13] show high-probability continuations are less diverse than human text and propose nucleus sampling, but the pressure toward sameness lives in the learned distribution, not only in sampling, which this study measures directly. Bommasani et al. [6] add scale: when one foundation model underlies millions of applications, its biases become the field's, making diversity a population-level concern.

3.2 Alignment and Output Diversity

Alignment adds a second layer on top. Kirk et al. [17] show that RLHF reduces output diversity compared to SFT, across syntactic, semantic, and logical dimensions, within one model family and only within-model. This study extends their result across architectures and developers, across a domain boundary, and down to the per-prompt level, the societal side Kirk et al. raise but do not measure. Yang et al. [27] measure the mechanism: alignment cuts the model's effective branching factor (usable next-token continuations) by two to five times, narrowing the output space from the first token. This study tests whether that narrowing is large enough to produce a measurable gap against human text. Santurkar et al. [23] show that this narrowing is not neutral: on subjective questions, aligned models lean toward the opinions of particular demographic groups instead of covering the range of views. That is the substantive side of the statistical narrowing, and it falls on the same advice and opinion questions where this study finds convergence strongest.

Read together, these two bodies describe *compounding* pressures: maximum-likelihood training already concentrates probability on dominant patterns, and preference alignment adds a second contraction on top of it. What neither line settles is whether the combined effect is strong enough that independent models from different developers converge on the *same* answer to a given prompt. This study takes up that question.

3.3 Empirical Evidence: Effects on Human Output Diversity

Padmakumar and He [20] ran the closest prior test: in a controlled co-writing experiment, essays written with InstructGPT, but not with the unaligned base GPT-3, showed a significant drop in content diversity and higher similarity between authors. That instruct-versus-base split is the same contrast this study draws (RQ3). Here it appears in the models' own outputs rather than in the human writing they help produce. At the group level, Ashkinaze et al. [3] and Anderson et al. [1] (the latter for creative ideation) find that diversity falls across a group of users even when no individual looks constrained. This study measures that dynamic one step upstream. Jain et al. [15] argue the stakes are task-dependent: output homogenisation is benign where one answer is correct, but harmful where the value lies in functional diversity, and they give a task taxonomy to separate the two cases. This study tests where convergence actually happens and finds a more mixed picture than either view predicts.

3.4 Approaches to Preserving Diversity

Two lines of work try to keep outputs diverse, at different points in the pipeline. At *decoding time*, truncated-sampling methods such as nucleus sampling [13], and raising the sampling temperature, widen the set of tokens a model may emit. These act on the output distribution after training, so they can only redistribute probability the model has already learned, not change the distribution itself. Consistent with this, the gap measured here narrows but does not close as temperature rises (Appendix H), which points to the convergence being a property of the learned distribution rather than of the sampling step.

At *training time*, pluralistic alignment [25] targets the cause more directly: rather than optimising toward the single response a majority of raters prefer, it aims to represent the spread of defensible positions and preserve the distribution of acceptable answers instead of collapsing it to its mode. This is the more promising direction, since the pressure toward sameness originates in the preference objective (Section 2). It has not yet been evaluated against the per-prompt, cross-model convergence measured here. The diagnostic introduced in this study is designed to fill that gap (Figure 2).

3.5 Research Gap

Prior work is incomplete at three joints. Kirk et al. [17] and Yang et al. [27] establish the mechanism within a single model family, not across organisations or domains. Padmakumar and He [20] come closest on attribution, isolating alignment with an instruct-versus-base contrast, but they measure how much the human essays written with one model differ from each other, not how far independent models' own answers converge. Ashkinaze et al. [3] and Jain et al. [15] motivate the downstream stakes at the group or topic level, not the individual response.

No prior work combines all three pieces this study requires: a controlled attribution of the diversity gap to alignment rather than architecture or training data; a test across domains including creative writing, where convergence is hardest to dismiss; and a link from corpus-level statistics to what an individual user actually receives. Each depends on the others: attribution without domain breadth cannot show the gap is general; domain breadth without attribution cannot tie it to alignment; and corpus-level results without a per-prompt link say nothing about what one user encounters.

4 Methodology

To isolate the effect of alignment from confounds such as topic and prompt difficulty, all models are compared on identical prompts.

4.1 Dataset and Generated Responses

The human baseline comes from HC3 [12], using its open-domain QA and finance subsets, and from Reddit WritingPrompts [9] for creative writing. Up to 500 prompt–response pairs are sampled from each domain. The human answers are crowd-sourced internet text, so the comparison is against the text aligned models displace in practice, not professionally edited writing (Section 6.3).

Four aligned models are queried on the same prompts at $T = 0.7$, capped at 300 tokens: Llama-3.1-8B-Instruct [7], Qwen3-32B-Instruct [21], Gemma-3-27B-it [11], and Apertus-8B-Instruct [2]. All four are generated under identical settings, so any gap against the human baseline is not a generation artefact. Together they span four organisations and three preference-optimisation algorithms (RLHF, DPO/GRPO, QRPO), which tests whether the gap generalises across alignment methods. ChatGPT responses come directly from HC3 (2023

GPT-3.5 outputs generated with an unknown temperature and prompt format) and are treated as a historical reference point rather than a co-equal controlled model. The RQ1/RQ2 conclusions rest on the four controlled models (Llama, Qwen3, Gemma, and Apertus). RQ4 and the style analysis instead use one model per developer (ChatGPT, Llama, Qwen3, and Gemma), omitting Apertus; RQ3 uses the Llama-3.1-8B and Mistral-7B instruct/base pairs.

For RQ3, two unaligned base models are compared against their own instruct versions: Llama-3.1-8B across all three domains and Mistral-7B-v0.1 [16] across the two knowledge domains. Because base and instruct share the same architecture and pre-training data, any difference between them is attributable to the post-training pipeline (SFT and preference optimisation, which cannot be separated here). Base models use a plain Q:/A: completion prompt. A few-shot prompt raises the share of base answers passing the 20-token filter but clusters responses around the exemplar, lowering measured base diversity and biasing the instruct-versus-base comparison, so the unprimed format is used despite the lower yield.

Apertus is the smallest and only fully-open model in the set and acts as a cross-organisation generality check across all three domains. In creative writing it tends to produce meta-commentary on the prompt rather than stories. This is reported but treated as task refusal rather than a diversity signal (Section 5.1). Full generation settings, the 20-token minimum filter, the 300-token truncation, and the chain-of-thought suppression flag used for Qwen3 are detailed in Appendix D. Table 7 (Appendix K) traces each sample size through every filtering step so each reported n can be matched to its analysis.

Response length. In open-domain QA, human responses are much shorter than AI responses (Appendix K, Table 9). A truncation-controlled re-analysis (Appendix G) shows the QA CR and LD gaps are strongly length-driven, so they are read as upper bounds, while SD is length-robust and carries this domain's conclusions. Lengths are comparable in finance and longer for humans in writing, so no length inflation applies there.

4.2 Diversity Metrics

No single measure captures diversity [26], so following Tevet and Berant [26] and Shaib et al. [24] we use three complementary metrics, one per family: semantic meaning, lexical richness, and structural redundancy. Each family has a different blind spot (embeddings can share biases with the models measured, type-token ratio sees only vocabulary, compression only repetition), so agreement across all three is more informative than any one alone. All three score the generated text, not the human viewpoints behind it. The gap between the two is a limitation (Section 6.3).

Semantic Diversity (SD). Each response is encoded with `all-MiniLM-L6-v2` [22]. Its SD score is the mean cosine distance to all other responses in its group G

(up to 500 responses from a single source; Section 4.1):

$$s_i \;=\; \frac{1}{|G|-1}\sum_{\substack{j\in G\\ j\neq i}}\bigl(1-\cos(e_i,\,e_j)\bigr)$$

where e_i, e_j are response embeddings, so $1-\cos(e_i, e_j)$ is 0 for identical meaning and larger for more distant meaning. SD is corpus-level: it measures spread *across the domain's prompts*, so a lower AI SD means tighter clustering across the corpus, not identical answers to any single question.

Lexical Diversity (LD). Vocabulary richness is the Type-Token Ratio (unique words over total words), averaged over a sliding 100-token window per response [24]. The window controls for length, which lowers raw TTR in longer texts.

Compression Ratio (CR). Following Shaib et al. [24], $\mathrm{CR}(r) = \mathrm{size}(\mathrm{gzip}(r))/\mathrm{size}(r)$ in UTF-8 bytes: gzip replaces repeated substrings with short back-references, so text that reuses the same words and phrases compresses far more than text where every passage is new. A lower CR means more repetition.

Effective number of distinct responses (Vendi Score). SD, LD, and CR give one score per response, which is what the paired tests need. As a single group-level summary, the Vendi Score [10] is also reported (full results in Appendix J):

$$\mathrm{VS} \;=\; \exp\left(-\sum_{i=1}^{n}\lambda_i\log\lambda_i\right)$$

where n is the group size, $\lambda_1, \ldots, \lambda_n$ are the eigenvalues (scaled to sum to one) of the normalised similarity matrix K/n, with $K_{ij} = \cos(e_i, e_j)$ the cosine similarity between responses i and j on the same embeddings used for SD. In plain terms it is the effective number of distinct responses in a set: near-identical answers score low, genuinely different ones score high. Groups are subsampled to a common size per domain before scoring, since the score grows with set size.

4.3 Statistical Design

Each test compares the human baseline against one aligned model, within one domain and metric, using the per-prompt scores as paired observations. The headline claim rests on Cohen's d holding the same sign and rough magnitude across these tests (metrics, models, and domains), not on any single comparison.

Each comparison gives up to 500 paired scores (one per prompt per group). These are tested with a **Wilcoxon signed-rank test**, a paired test that ranks the per-prompt score differences and asks whether they lean systematically in one direction, without assuming a normal distribution. Significance is set at $\alpha = 0.05$, with **Benjamini-Hochberg FDR correction** [5] across all pairwise comparisons, which adjusts each p-value to control the false discovery rate (the share of false positives) when many tests run at once. Effect size is reported as **Cohen's** d: the difference between the two groups' mean scores divided by their pooled (combined) standard deviation, so that $d = 1$ means the means lie one standard deviation apart. By convention $|d| > 0.8$ is large. Each d is reported with a 95% confidence interval from a paired bootstrap (2000 resamples); full intervals are in the results files.

4.4 Per-Prompt Convergence Analysis (RQ4)

The corpus-level metrics measure spread across the whole dataset. RQ4 asks a narrower question: for any given prompt, do aligned models answer more like each other than like the human?

For each prompt where all selected models and the human baseline have a valid response, all pairwise cosine distances are computed and split into two groups. AI–AI pairs are distances among the aligned models' answers. Human–AI pairs are distances between the human answer and each AI answer. A prompt is *convergent* if its mean AI–AI distance is smaller than its mean Human–AI distance.

The model set for this analysis is: ChatGPT, Llama-3.1-8B-Instruct, Qwen3-Instruct, and Gemma-3-27B-it in the two knowledge domains ($\binom{4}{2} = 6$ AI–AI pairs per prompt); and Llama, Qwen3, and Gemma in writing ($\binom{3}{2} = 3$ pairs), since HC3 writing has no ChatGPT data. Apertus is omitted here: queried independently as the generality check, including it would shrink the all-models prompt intersection and conflate that check with the convergence test.

Three features bound the interpretation. First, the knowledge-domain AI–AI pairs include ChatGPT, the one uncontrolled model, so some distances mix generation-config differences with convergence. Writing is a fully controlled three-model test, and a controlled-only re-analysis (Section 5.3) shows the convergence strengthens once ChatGPT is excluded, so it is not driving the result. Second, HC3 keeps one human response per prompt, so the basic test contrasts AI–AI with Human–AI agreement. Where a corpus supplies several human answers (finance via FiQA, writing via Reddit WritingPrompts), a Human–Human baseline is also computed, leaving only open-domain QA on the single-human comparison. Third, cosine distance captures *register* (how a response is written, as opposed to what it says) alongside content, so convergence here is convergence in style and substance combined. Each domain reports the mean distances, the Wilcoxon test, Cohen's d, and the percentage of convergent prompts. Stated as a reusable procedure, this is the diagnostic of Appendix A (Figure 2).

4.5 Style Feature Analysis

To identify what aligned models converge *toward*, each response is scored on six interpretable style features that together stand in for its register: length, hedging markers, balancing connectives, list and header markers, second-person address, and mean sentence length. Four aligned models (ChatGPT, Llama, Qwen3, and Gemma, one per developer) are compared against the human baseline per domain. Apertus is excluded here for the same reason as in Section 4.4. Per-feature bootstrap 95% confidence intervals on the human-minus-AI effect are used to assess whether each feature differs reliably across all four models.

4.6 Encoder Robustness and Implementation

To rule out encoder-specific bias, the SD analysis is repeated with `paraphrase-mpnet-base-v2` [22], an independently trained encoder (Appendix F). The full software stack and implementation details are given in Appendix D, and all code is available in the project repository.

5 Results

All results use up to 500 matched prompt pairs per domain after the 20-token filter. "Human" refers to HC3 and WritingPrompts (crowd-sourced internet text). ChatGPT rows are the uncontrolled HC3 reference (2023 GPT-3.5) and read as historical context. The RQ1/RQ2 conclusions rest on the four controlled models. Table 1 reports Cohen's d ($|d| > 0.8$ large; all $p < 0.001$ BH-corrected unless marked). Positive d: humans more diverse. Figure 6 visualises all effect sizes; Appendix K (Table 8) gives raw means. Table 6 (Appendix K) summarises each research question's answer and the strength of its evidence.

5.1 The Diversity Gap and Its Generality (RQ1, RQ2)

In both knowledge-based domains, humans are more diverse than every aligned model on **semantic diversity** (SD). SD gaps survive length-truncation to 20 words (Table 4), confirming they are not a length artefact. The open-domain QA conclusion rests on this length-controlled SD rather than raw scores. The gap holds across all five models (ChatGPT, Llama, Qwen3, Gemma, Apertus), spanning different architectures, developers, and release dates, and never reverses in finance. Apertus, the smallest model, shows the weakest effect at $d = 0.20$–0.30. CR and LD point the same way but are inflated by the human/AI length asymmetry in open-domain QA (‡; Appendix G), so they are reported as upper bounds there. RQ1 and the knowledge-based half of RQ2 are answered affirmatively.

Creative writing is model-dependent rather than uniformly reversed (HC3 has no ChatGPT stories). On SD, Llama is *more* diverse than humans, Qwen3 is indistinguishable, and Gemma is *less* diverse; Apertus tops the writing SD because it produces meta-commentary rather than stories, so its score marks task refusal. On LD and CR all three controlled models move the other way, more varied in vocabulary and less compressible than humans, so surface richness and meaning come apart. RQ2 is therefore partially answered: the gap replicates cleanly in knowledge-based domains but is mixed in writing, which is informative rather than inconclusive. A large writing SD can mark refusal rather than creativity (Appendix O). Table 1 gives the full effect sizes for every comparison; Figure 6 (Appendix I) visualises them.

Table 1. Cohen's d for all pairwise comparisons (Wilcoxon signed-rank, BH-corrected). †: not significant ($p_{\mathrm{fdr}} \geq 0.05$). ‡: open-domain QA LD/CR values inflated by the human/AI length asymmetry, read as upper bounds (Appendix G; the same sweep confirms SD is length-robust). Positive d: humans more diverse; negative: higher diversity in group b. SD: semantic diversity; LD: lexical diversity; CR: compression ratio. RQ1/RQ2 are human vs. aligned model; RQ3 is instruct vs. base.

Comparison	Domain	SD	LD	CR
RQ1 / RQ2: Human vs. aligned model				
vs. ChatGPT	open_qa	0.77	$1.58^{\ddagger}$	$1.82^{\ddagger}$
vs. ChatGPT	finance	0.17	1.50	1.50
vs. Llama-3.1 Inst.	open_qa	0.54	$1.22^{\ddagger}$	$1.85^{\ddagger}$
vs. Llama-3.1 Inst.	finance	0.62	0.84	1.55
vs. Llama-3.1 Inst.	writing	−0.55	0.24	$-0.03^{\dagger}$
vs. Qwen3 Inst.	open_qa	1.67	$0.95^{\ddagger}$	$1.97^{\ddagger}$
vs. Qwen3 Inst.	finance	1.22	$0.20^{\dagger}$	0.78
vs. Qwen3 Inst.	writing	$0.03^{\dagger}$	−0.34	−0.97
vs. Gemma-3-27B-it	open_qa	1.72	$0.67^{\ddagger}$	$1.89^{\ddagger}$
vs. Gemma-3-27B-it	finance	1.05	−0.18	0.71
vs. Gemma-3-27B-it	writing	1.26	−1.17	−1.61
vs. Apertus-Inst.	open_qa	0.20	$0.93^{\ddagger}$	$1.25^{\ddagger}$
vs. Apertus-Inst.	finance	0.29	0.91	1.27
RQ3: instruct vs. base (negative d: instruct less diverse than base)				
Llama-3.1-8B	open_qa	−0.43	0.35	−0.45
Llama-3.1-8B	finance	−1.25	0.72	−0.47
Llama-3.1-8B	writing	−2.12	$0.11^{\dagger}$	−0.82
Mistral-7B	open_qa	−0.23	$-0.16^{\dagger}$	−0.88
Mistral-7B	finance	−0.95	$0.18^{\dagger}$	−0.97

5.2 Alignment Attribution (RQ3)

RQ1 establishes the gap exists across models in the knowledge-based domains. RQ3 asks whether alignment contributes to it, or whether it is an artefact of model capacity or training data, by comparing an instruct model against its own unaligned base, where architecture and pre-training are held constant so post-training alignment is the only difference. To test whether the effect is family-specific, this is done for two independent families: **Llama-3.1-8B-Instruct** vs. **Llama-3.1-8B** (all three domains), and **Mistral-7B-Instruct** vs. **Mistral-7B** (the two knowledge domains; base writing continuations are degenerate, as below).

Llama-Instruct is less diverse in meaning than Llama-Base in all three domains, most strongly in writing, and CR follows the same direction (Table 1, RQ3 rows). Vocabulary runs the other way in the knowledge-based domains, where Llama-Instruct has *higher* LD than the base (taken up in Section 6). Since architecture and pre-training are held constant, the SD and CR gap is most simply attributed to the post-training alignment pipeline, though this comparison cannot separate SFT from preference optimisation. The open-domain QA SD cell is the weakest (faint for Llama at $d = -0.43$, fainter for Mistral at $d = -0.23$), so the RQ3 conclusion rests on the finance and writing SD gaps and the CR gaps, which are large and replicate across both families.

The Mistral family replicates the core effect: Mistral-Instruct is less diverse in meaning (SD) and more compressible (CR) than its own base in both knowledge domains, finance most strongly (SD $d = -0.95$ $[-1.06, -0.85]$; CR $d = -0.97$) and open-domain QA more faintly ($d = -0.23$ $[-0.38, -0.10]$; $n = 151$), the same direction as Llama. The SD and CR reductions thus hold across two independent families. Vocabulary is the exception: Mistral's LD effect is null, so the LD enrichment seen in Llama is family-specific rather than a general alignment effect. Writing is striking but bounded: Llama-Instruct shows no corpus deficit against humans in writing yet compresses most strongly against its own base, as alignment narrows hardest where the starting space is widest. Base writing continuations are noisier, however, so part of the base's diversity may reflect incoherence rather than usable variety [17].

Disentangling alignment from shared data. A natural objection is that models from different developers share pre-training data and partial cross-distillation, so they might converge for reasons unrelated to alignment. The instruct/base pairs test this at the cross-family level: we compute per-prompt convergence *between* the two families' base models (Llama-3.1-8B base vs. Mistral-7B base) and compare it with the same measure between their instruct versions. In open-domain QA the two base models do *not* converge more than humans (54.6% of prompts, $d = 0.10$, $p = 0.23$). Aligning them yields strong cross-family convergence (86.9%, $d = 0.94$), so here the effect is alignment, not shared data. In finance the base models partially converge (78.7%, $d = 0.89$), which confirms a shared-data contribution. Yet alignment more than doubles it (99.6%, $d = 2.50$) and more than halves the pairwise distance (0.37 to 0.17). Alignment therefore appears to be the main driver of cross-developer convergence in both domains, clearest in QA and dominant in finance. This is a two-model probe (the two families with a public base), so it is read as directional evidence rather than a full controlled comparison. The QA base comparison also rests on the 130 prompts where both base completions clear the 20-token filter, plausibly the more coherent ones, so it is the weaker of the two domains here.

5.3 Per-Prompt Convergence (RQ4)

The societal concern is narrower than corpus-level spread: if many users ask the same question, do they receive the same answer? Following Section 4.4, if AI systems converge, AI–AI distance should be consistently smaller than Human–AI distance on the same prompt.

Using the AI–AI pairing of Section 4.4, all three domains show the predicted pattern: AI–AI distance is consistently smaller than Human–AI distance, with large effect sizes (Cohen's d from 1.01, 95% CI $[0.87, 1.15]$, in open-domain QA to 2.28 $[2.12, 2.47]$ in finance) and 87–98% of prompts convergent in every domain (all $p < 0.001$, Wilcoxon; Figure 1).

Robustness to the uncontrolled model. The headline does not depend on ChatGPT, the one uncontrolled model (Section 4.1). Re-running the two knowledge-based domains on the three controlled models only (Llama, Qwen3, Gemma; writing already excludes ChatGPT) makes the convergence *stronger*, not weaker: open-domain QA rises from 87.3 to 91.4% of prompts convergent (d from 1.01 to 1.18) and finance from 98.1 to 99.4% (d from 2.28 to 2.46). The older ChatGPT answers sit slightly apart from the newer models, so including them loosens the AI cluster. The controlled models converge more tightly still. The headline is therefore a property of the controlled comparison, not an artefact of the uncontrolled baseline.

Corpus spread and length. The per-prompt result does not contradict the absent corpus-level writing gap (Section 5.1): corpus SD measures spread *across* prompts, per-prompt distance *within* one. Models vary across prompts but converge on any single one, answering more like each other than like a human. The result is not a length artefact. Re-running the per-prompt distances after truncating every response to a common word budget (Appendix C, Table 2) leaves the convergence intact at every budget down to 40 words, with the gap large and significant in all three domains (Cohen's d from 0.82 in open-domain QA to 2.32 in finance at $L = 40$, and 79–98% of prompts convergent). This tracks the length-robust behaviour of corpus SD (Table 4) rather than the length-sensitive CR and LD.

A Human–Human baseline. Comparing the human answers to one another confirms the convergence is not merely distance from one idiosyncratic human. Where a corpus supplies several independent human answers to the same prompt, the per-prompt distance can be computed *among* humans, the cleaner test of whether aligned models converge beyond the natural spread of human answers. This paired test is available in two domains, including the headline one. Open-domain QA has no matched multi-answer corpus, so of the three it is the least firmly anchored and rests on the external reference below. In **finance**, HC3 draws its questions from FiQA, which keeps several human answers per question even though HC3 itself records only one. On the 270 finance prompts with at least two human answers, AI–AI distance is smaller than Human–Human on 97.8% of them (0.20 vs. 0.47; Cohen's $d = 2.74$). In **writing**, Reddit WritingPrompts supplies several human stories per prompt: on the 384 prompts with at least two, AI–AI distance is smaller on 94.3% (0.43 vs. 0.64; $d = 2.05$). In both domains, and most strongly in finance where the monoculture concern centres, the aligned models cluster more tightly than independent humans do (Figure 1, whose panels use this multi-answer subset, 270 in finance and 384 in writing, not the full intersection used above).[1]

A negative control. The same Human–Human baseline also checks the diagnostic of Figure 2: in finance the Human–Human distance (0.47) is statistically indistinguishable from the Human–AI reference (0.46), so the human group is not flagged, whereas the AI group is flagged decisively against that same reference (AI–AI 0.20; $d = 2.74$). The flag fires on the AI cluster's abnormal tightness and stays quiet on the normal spread of human answers, as a usable diagnostic requires.

[1] These gaps are cosine-*distance* ratios, read for sign and size rather than as linear similarity claims.

Open-domain QA. Here no matched multi-answer corpus aligns with our prompts, so the paired test gives way to an external reference: on ELI5 explanatory questions [8] ($n = 371$), Human–Human distance averages 0.48, well above the QA AI–AI distance (0.18). This comparison is unpaired, but with paired confirmation in finance and writing, the single-human QA result is unlikely to be an artefact of one atypical human.

A within-model floor. A final alternative is that the AI–AI tightness merely reflects low per-model entropy: resampling one aligned model five times per prompt gives a self-distance below the cross-model AI–AI distance in every domain, with AI–AI exceeding the floor on 88–92% of prompts ($d = 0.98$–1.12; Table 11). Different developers' models thus differ from one another by more than one model differs from itself, yet cluster far below the human distances. The floor is measured on a single model, so it is indicative rather than an exact per-model bound (Appendix L), but it shows the cross-developer tightness is more than a sampling-band artefact.

Register and substance. The measure captures the two combined. Stripping markdown and re-embedding leaves convergence unchanged (Section 5.4), ruling out shared formatting. Whether the clustered answers also make the same argument is harder. A manual coding of 20 high-convergence finance prompts (Appendix K) finds the convergence substantive on most: ten of the twelve most convergent prompts, and six of the eight advice-and-opinion prompts (the cell carrying the monoculture claim), coincide in content rather than voice alone. On whether to keep student debt, for instance, all three models raise the same tax-deduction and opportunity-cost arguments rather than spanning the range a diverse set of advisers would. The convergence is therefore substantive more often than not, including where it matters most, though the metric can overstate it; register and content are reported together, not claimed to be separated.

5.4 Robustness Check

Three checks guard the main findings. First, repeating the SD analysis with an independently trained second encoder gives mean SD scores that correlate at Pearson $r = 0.92$ ($p < 0.001$), with no comparison changing direction or losing significance (Appendix F). Second, stripping all markdown (headers, bullets, emphasis) and re-embedding leaves per-prompt convergence essentially unchanged (stripped vs. original): 88.7 vs. 87.3% of prompts convergent in open-domain QA, 98.3 vs. 98.1% in finance, and 94.6 vs. 95.0% in writing, with Cohen's d shifting by at most 0.07. Third, a group-level Vendi Score [10], using neither pairwise distance nor compression, reproduces the same ranking (humans highest in both knowledge domains, every aligned model lower, writing model-dependent; Appendix J). The convergence is a property of the content, not formatting, and agreement across four structurally different metrics makes a measurement artefact unlikely.

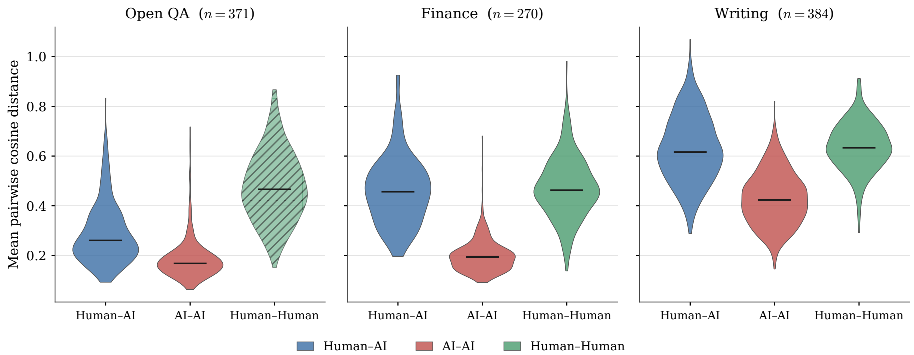

Fig. 1. Per-prompt pairwise cosine distances by domain. Each violin shows the distribution, across prompts, of the mean pairwise distance within a comparison: Human–AI (blue), AI–AI (red), and Human–Human (green), with the median marked. Finance and writing Human–Human are prompt-matched (FiQA, $n = 270$; WritingPrompts, $n = 384$; prompts with ≥ 2 human answers), so the AI and human distributions are measured on the same questions. The QA Human–Human is an external reference from ELI5 [8] ($n = 371$, hatched), since HC3 supplies a single human answer per QA prompt and admits no within-prompt human pairs. In every domain the AI–AI distribution lies below both human baselines: AI–AI is smaller than Human–AI on 87–98% of prompts, and smaller than the prompt-matched Human–Human on 97.8% of finance and 94.3% of writing prompts (QA AI–AI 0.18 versus the ELI5 reference 0.48). Because the distances are cross-model, two users querying different developers' models receive answers closer to each other than to a human.

5.5 The Shape of the Convergence

Following Section 4.5, open-domain QA shows a clear shared register: aligned answers are longer, more hedged, use more balancing connectives, address the reader more, and use shorter sentences, with every feature's 95% confidence interval excluding zero and all four models drifting the same way on five of the six (Figure 4). The length-normalised features carry the result, and their shared direction across four independent models does not arise by chance (per-feature bootstrap intervals and the sign test in Appendix E). The pattern weakens to longer-and-more-hedged in finance and is absent in writing, tracking the model-dependent corpus pattern of Section 5.1.

6 Discussion

6.1 Interpretation of Results

Per-prompt convergence. On 87–98% of prompts, aligned models answer more like each other than like the human, including in writing where the corpus gap is absent. Because the distances are cross-model, between four independent developers, whichever aligned model a user reaches for returns an answer from the same narrow region.

Convergence across question types. It holds in both: 98.3% ($d = 2.55$) on questions a lexical rule tags as advice-and-opinion and 95.7% ($d = 1.66$) on those it tags as factual. A 30-prompt hand-check shows that rule is coarse (67% agreement with human labels, Cohen's $\kappa = 0.26$, mostly by under-tagging intent-based advice as factual), but that cuts the right way: even a factual bucket contaminated with real advice still converges at 95.7%, so the conclusion that convergence does not spare the open-ended questions the monoculture concern is about does not rest on a clean split.

Two kinds of diversity in writing. The corpus gap is inconsistent in writing because alignment settles on preferred story framings while raters disagree on what makes a good story; yet per-prompt convergence stays strong, so statistical novelty is not narrative originality. In the appendix example, an aligned model's story scores as statistically novel yet reads as narratively empty, while the statistically ordinary human story is the genuinely creative one (Appendix O). This is the clearest case where the metric and the quality it stands in for come apart.

Why the effect points to alignment. Two parts of the evidence do different jobs. The cross-developer comparison establishes *direction*: the gap repeats with the same sign across all five models and both knowledge domains, even as its magnitude varies ($d = 0.20$–1.72 in open-domain QA). The controlled instruct-versus-base comparison isolates *attribution* at the 8B scale: switching alignment on within a family produces the gap, in two independent families. With alignment off, the base models do not converge in QA and only partially in finance, so the cross-developer effect is better explained by alignment than by shared pre-training data (Section 5.2). Neither alone suffices. Together they make the magnitude spread expected rather than troubling: architecture, scale, and data differ across developers, shifting effect sizes without changing their sign. Alternatives to alignment are weighed elsewhere: shared pre-training data is a partial contributor in finance only (Section 5.2), low per-model entropy is ruled out by the within-model floor, and model scale remains a residual confound since the controlled test runs at 8B (Section 6.3).

6.2 Societal Implications

The convergence chain. Going from text-level convergence to a claim about collective intellectual diversity needs two further steps: users must be exposed to the same AI outputs, and those outputs must shape their thinking. The per-prompt result (Section 5.3) supports the first step. For the second, Ashkinaze et al. [3] give experimental evidence that high AI exposure reduces the diversity of ideas across a group while leaving individual quality unchanged. Neither condition alone is sufficient: converged text is harmless without transfer, and transfer is harmless without converged text. This is an inference about a mechanism, not a measured downstream harm.

Some convergence is benign. In medical, tax, or safety information, convergence is a *feature*: users want consistent ground truth, not conflicting framings. The concern is selective, applying where value lies in the range of perspectives (investment, career, creative decisions) and an aligned model delivers one framing where a diverse adviser pool would offer several. Finance sits squarely here and shows the strongest convergence of the three domains [15].

Mitigation. Widening decoding through higher temperature or nucleus sampling appears insufficient: in the check we ran (one model, one domain; Appendix H) the gap narrows but does not close as temperature rises, which suggests the convergence sits in the learned distribution rather than the sampling step. That leaves the alignment objective as the lever, where pluralistic alignment [25] is the natural candidate and the per-prompt diagnostic (Figure 2) is the test of whether it works: a successful method should lift AI–AI distance toward the human baseline on the same prompts. The diagnostic assembles standard parts (sentence embeddings, cosine distance, paired tests); its contribution is the framing, applied per prompt across independent models. Used as a screen it is informative, but optimised against directly it invites Goodhart's law, since a model can raise AI–AI distance without diversifying content, so any gain should be confirmed on substance (Section 5.3), not on the metric alone.

6.3 Limitations

Attribution within the pipeline. The within-family comparisons (RQ3, two families) and cross-model replication implicate the post-training pipeline as the most plausible common cause, but cannot isolate which stage is responsible. The alignment signal (RLHF or DPO) is the most direct route to narrower output and the one the evidence most supports, but SFT runs in sequence with preference optimisation and cannot be separated here. Prompt format is a residual confound: base models receive completion-style prompts and instruct models chat-formatted ones. The non-Llama models also differ in architecture, size, and pre-training data, plausibly driving the differing effect sizes (d = 0.20–1.72 in open-domain QA) even though the shared direction points to alignment. Single-family studies such as Kirk et al. [17] gain cleaner attribution by holding architecture and data fixed within one model; this study trades some of that precision for breadth across organisations, and the two designs are complementary. Shared pre-training data is a further candidate, but the base-model comparison (Section 5.2) shows it contributes in finance and not in open-domain QA, so it is at most a partial explanation.

Scale of the attribution. The attribution rests on 8B-scale comparisons, whereas the largest corpus gaps come from the 27–32B models. A Llama-3.1-70B base/instruct check, run identically, finds the semantic-diversity reduction far weaker at this larger scale within the same family (finance Cohen's $d = -0.17$ vs. -1.25 at 8B; open-domain QA near zero; Appendix M). That comparison is itself confounded, since a 70B base produces coherent, varied completions that inflate base diversity, so whether the effect attenuates with scale or the confound masks it is unclear. The generality of the attribution across scales is therefore tested here, and the test is inconclusive.

Scope and settings. The $T = 0.7$ design is the moderate production setting; the gap holds at $T = 0.3$ and 1.0 (Appendix H), but a full sweep is future work. HC3 is from 2023 (GPT-3.5 era) and English-only, so the baseline is period- and language-specific, and the first human response per prompt is used where HC3 offers several, making the reported gap a lower bound. The three domains are illustrative, not exhaustive: high-stakes settings such as medicine and law, and non-Transformer architectures, are untested and may differ.

Variability versus value. Reddit and Q&A-forum answers are self-selected, so the gap is best read as aligned models vs. crowd-sourced internet text rather than AI vs. human in general. The diversity metrics reward *variance*, and forum text varies through incoherence and low effort, not only through valuable plurality. To bound this, we hand-coded the twenty most dispersed human finance prompts (Appendix N): thirteen show genuine plurality of defensible positions, six are mixed, and one is a low-effort non-answer. Low-effort text inflates the gap but does not explain it; on the prompts most at risk the dispersion is mostly real disagreement, not noise.

Text versus thought. This study measures text diversity, not diversity of thought. The downstream link rests on one outside study [3] and remains a research question.

Metrics versus perspective. The cosine metric captures register and substance combined. A paired Human–Human baseline is in hand for finance and writing; only open-domain QA still lacks a matched multi-answer corpus and leans on an external ELI5 reference. The dual-encoder check guards the SD findings against encoder bias (Appendix F).

Apertus. Apertus-8B-Instruct is a cross-organisation generality check rather than a co-equal peer: it is the smallest model and its writing responses are meta-commentary rather than stories, marking task refusal (so it has no writing SD cell in Table 1). It also shows the *weakest* knowledge-domain gap in the set (SD $d = 0.20$–0.30). We read this as a boundary case that cuts both ways: it is consistent with a dose-response, where the least aggressively aligned model converges least, but a smaller model also has less to compress, so Apertus is not strong evidence either way. The core claims hold without it.

7 Conclusion

This study tested whether preference-aligned language models produce less diverse text than humans, and whether that narrowing traces to the alignment step. The evidence gives a clear answer for information-seeking.

In open-domain QA and finance, humans are measurably more diverse than aligned models across all five tested (Figure 6). Controlled instruct-versus-base comparisons in two families (Llama, Mistral) implicate the post-training pipeline: semantic- and compression-diversity reductions replicate in both families, though the vocabulary effect is family-specific. The sharpest evidence is the paired Human–Human test: where a corpus supplies several human answers per prompt, the aligned models cluster more tightly than independent humans do, on 97.8%

of finance prompts and 94.3% of writing prompts, strongest in finance, the domain the concern centres on (Section 5.3).

Output convergence is a necessary but not sufficient condition for the broader AI-induced monoculture risk. Whether it narrows diversity of thought is supported by Ashkinaze et al. [3] but not established here. The most direct technical response is pluralistic alignment [25], which preserves the spread of defensible answers rather than collapsing raters onto one preferred response. The per-prompt metric introduced here gives such methods a concrete target: a successful pluralistic model should shrink the AI–AI-below-human gap, though whether current methods do so is untested.

Three extensions would sharpen the picture: separating register from substance directly (the cosine metric combines them, and the manual coding only spot-checks their alignment); isolating which post-training stage drives the effect (SFT-only checkpoints) and a less confounded large-scale base/instruct comparison than the 70B check here; and a matched Human–Human baseline for open-domain QA, the one domain still on a single-human comparison.

The actionable contribution is the diagnostic itself (Figure 2): a cheap pre-deployment check that flags domains where models cluster more tightly than a human or reference baseline. Reported in model cards, it would catch the per-prompt convergence that corpus-level evaluations miss. More broadly, the AI tools millions rely on for open-ended advice may be narrowing the range of framings people encounter, a measurable and fixable risk.

8 Ethical Considerations

This study uses only public datasets (HC3 and WritingPrompts) and public model APIs and weights; no new human-subjects data was collected, the “human” baselines are pre-existing anonymous crowd-sourced text, and only short excerpts are reproduced (Appendix O). The work is diagnostic, measuring a homogenisation risk to make it visible and fixable rather than to exploit it, and its societal claims are kept narrow (text-level diversity, not diversity of thought, with effect sizes rather than demonstrated harms; Section 6.3). Code and analysis scripts are available so the results can be checked and extended.

A The Per-Prompt Convergence Diagnostic

Figure 2 states the reusable pre-deployment check introduced in Section 4.4 as a four-step procedure.

Per-Prompt Convergence Diagnostic

1. Fix a prompt set for the target domain and collect one response per prompt from each of $k \geq 2$ deployed models, plus one human or reference answer per prompt.
2. Embed every response with a fixed sentence encoder; for each prompt, compute the mean pairwise AI–AI cosine distance and the mean Human–AI (or reference) distance.
3. Test the paired per-prompt differences with a Wilcoxon signed-rank test and report Cohen's d.
4. *Flag* the domain when AI–AI distance is smaller than the Human/reference distance on a majority of prompts with a large effect ($|d| > 0.8$, $p < 0.05$): the models converge on the same answers behind any corpus-level diversity.

Fig. 2. A reusable pre-deployment check for per-prompt convergence. It runs on a fixed prompt set without retraining and catches per-prompt sameness that corpus-level diversity metrics miss (Section 5.3).

B Conceptual Overview

Figure 3 (referenced from the Introduction) gives the full-size version of the pipeline diagram.

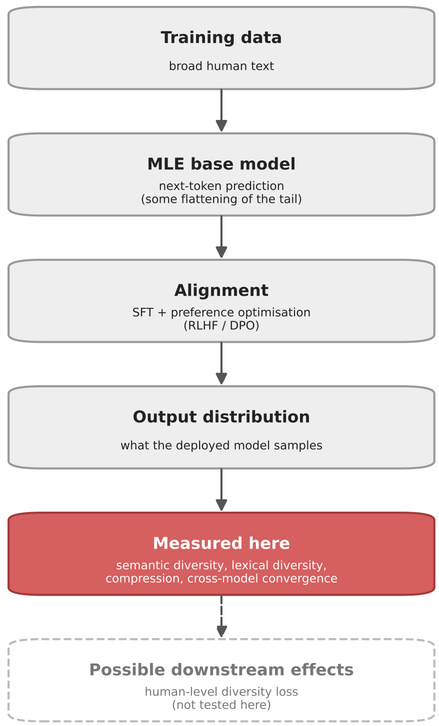

Fig. 3. Where this study sits in the training pipeline. Maximum-likelihood pre-training already flattens the tail of the human distribution; alignment (supervised fine-tuning followed by preference optimisation) is the additional step examined here. The measured quantities (highlighted) are properties of the deployed output distribution. Human-level diversity loss is a plausible downstream consequence but is not tested in this study.

C Per-Prompt Convergence: Length Control

The per-prompt convergence result (RQ4, Section 5.3) uses full-length responses in the main analysis. To confirm it is not driven by the human/AI length asymmetry, the per-prompt Human–AI and AI–AI cosine distances were recomputed after truncating every response to a common word budget L, using the same prompt intersections and the same procedure as Figure 1. Table 2 reports Cohen's d (Human–AI vs. AI–AI distance) and the percentage of convergent prompts at each budget. The convergence holds at every budget down to 40 words: the effect stays large ($d \geq 0.8$) and significant ($p < 0.001$, Wilcoxon) in all three domains, and the share of convergent prompts moves only a few points. This matches the length-robust behaviour of corpus SD (Table 4) and rules out length as an explanation for the headline finding.

Table 2. Length-controlled per-prompt convergence. For each domain and word budget L, Cohen's d compares mean Human–AI against mean AI–AI cosine distance (positive = AI answers closer to each other than to the human), and "% conv." is the share of prompts with mean Human–AI > mean AI–AI distance. L = none is the full-length analysis of Figure 1. All Wilcoxon tests $p < 0.001$. $n = 371$ (open-domain QA), 482 (finance), 464 (writing).

Domain	Statistic	L=40	L=70	L=100	L=none
Open-domain QA	Cohen's d	0.82	0.87	0.88	1.01
	% conv.	78.7	81.9	82.2	87.3
Finance	Cohen's d	2.32	2.37	2.33	2.28
	% conv.	97.9	98.1	98.3	98.1
Creative writing	Cohen's d	1.53	1.68	1.74	1.66
	% conv.	90.9	91.4	92.5	95.0

D Generation and Filtering Details

All models are queried at $T = 0.7$ with a 300-token maximum, via the Groq and Hugging Face Inference APIs, on the exact prompt sets used for the human baselines, so all comparisons within a domain share the same prompts. Qwen3 prompts include the `/no_think` directive to suppress chain-of-thought output. Responses shorter than 20 tokens are excluded as uninformative, and those over 300 tokens are truncated for length comparability. Failed generations are not regenerated, in line with the fixed-data design in which every model's data was collected once before any analysis.

All experiments are implemented in Python: HC3 is loaded via HuggingFace Datasets [18], responses are generated through the Groq and HuggingFace Inference APIs, embeddings are computed with `sentence-transformers`, statistical tests with `scipy`, and compression with `gzip`.

E Shape of the Convergence: Full Figure

Figure 4 gives the full per-feature effect sizes behind the shared-register result of Section 5.5. The inference rests on the per-feature bootstrap 95% confidence intervals (each excluding zero): a sign test over the six features gives $p = 0.0002$, but the features are positively correlated, so that p-value is anti-conservative and is reported only as a descriptive summary.

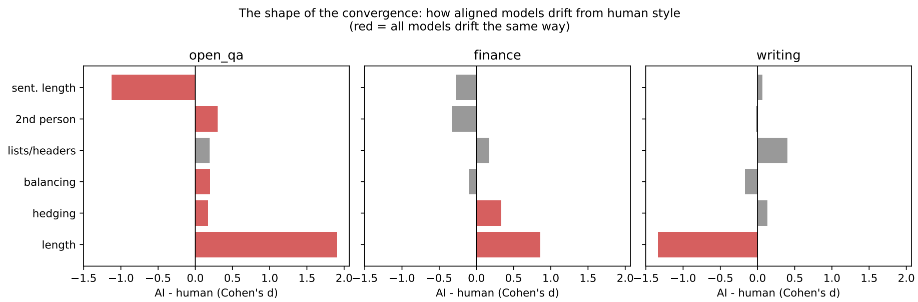

Fig. 4. The shape of the convergence. For each style feature, the bar is the AI-minus-human difference (Cohen's d, pooled aligned models; positive means the aligned models score higher). Red bars mark features where every model differs from humans in the same direction. The shared register is statistically clear in open-domain QA (five of six features, sign-test $p < 0.001$: longer, more hedged, more balancing connectives, more second-person, shorter sentences), weaker in finance (length and hedging only), and absent in writing.

F Robustness Check

The primary SD analysis uses `all-MiniLM-L6-v2` as the sentence encoder. To test whether the finding depends on encoder choice, the SD computation was repeated with `paraphrase-mpnet-base-v2`, an independently trained model. Figure 5 plots both encoders' mean SD scores for each group–domain combination. The Pearson correlation is $r = 0.92$ ($p < 0.001$). No comparison changes direction or loses significance after BH correction. The rank ordering of groups is stable across encoder choice.

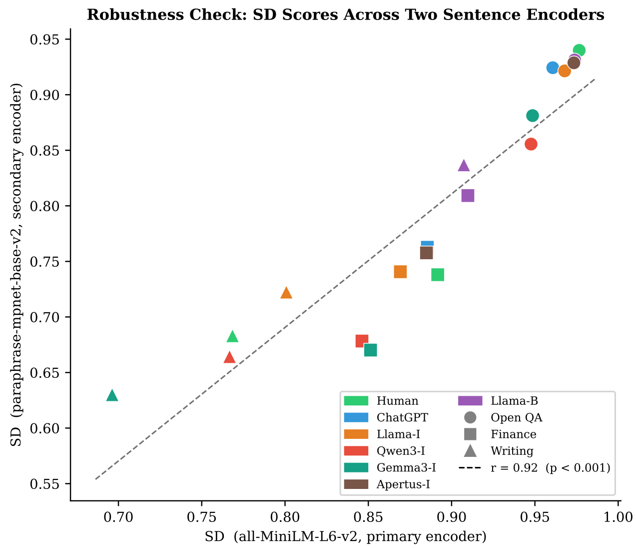

Fig. 5. Robustness check: mean SD per group–domain with the primary encoder (`all-MiniLM-L6-v2`, x-axis) versus the secondary encoder (`paraphrase-mpnet-base-v2`, y-axis). Each point is one group–domain combination; colours indicate model group, markers indicate domain. The tight linear relationship ($r = 0.92$) makes encoder-specific bias an unlikely driver of the results.

G Length-Control Robustness Check

Section 4 notes that human responses in open-domain QA are far shorter than AI responses (43 ± 24 words versus 112–168; Table 9). To test whether this asymmetry drives the open-domain QA gaps rather than a genuine diversity difference, the metrics are recomputed after truncating every response in both groups to a common word budget L, sweeping L from close to the human mean length up to no truncation. Compression (CR) and lexical diversity (LD) are reported first (Table 3); semantic diversity (SD), the metric the open-domain QA conclusion rests on most heavily, is swept separately afterward (Table 4). For every metric the $L =$ none column uses the same response-selection rule as the main analysis, so it reproduces the corresponding Table 1 value.

Every CR and LD effect in open-domain QA falls to near zero, or reverses sign, at $L = 20$–40 (close to the 37-word human mean), then grows steadily as the budget increases toward the unmatched full-length comparison in Table 1. This is the signature of a length artefact, not a genuine redundancy or vocabulary difference: forced to the same short length, most of the gap disappears. LD's windowed correction (Section 4) only controls for length once a response exceeds the 100-token window, and the 37-word human QA responses fall below it, so LD partly reduces to uncontrolled type-token ratio here.

Table 3. Cohen's d for the Human-vs-aligned-model gap in open-domain QA, recomputed after truncating every response to a common word budget L (words). L = none uses the same response-selection rule as Table 1 (the first response per prompt that passes the 20-token filter): CR at L = none reproduces the corresponding Table 1 cell exactly in all five rows. LD at L = none is within 0.00–0.09 of Table 1 (identical for ChatGPT, up to 0.09 for Gemma), because this sweep's tokenizer is case-sensitive while the main pipeline's lowercases before computing type-token ratio; this is a difference between the two scripts, not a sampling discrepancy, and it does not change which cells collapse under truncation. Positive d indicates humans more diverse (CR: less redundant; LD: more lexically varied).

Comparison	Metric	L=20	L=40	L=70	L=100	L=none
vs. ChatGPT	CR	−0.13	0.81	1.47	1.74	1.82
vs. ChatGPT	LD	0.05	0.60	1.30	1.64	1.58
vs. Llama-3.1 Inst.	CR	−0.05	0.66	1.22	1.51	1.85
vs. Llama-3.1 Inst.	LD	−0.18	0.25	0.83	1.20	1.21
vs. Qwen3 Inst.	CR	0.10	0.58	1.21	1.55	1.97
vs. Qwen3 Inst.	LD	−0.59	−0.25	0.35	0.79	0.89
vs. Gemma-3-27B-it	CR	−0.30	0.50	1.18	1.49	1.89
vs. Gemma-3-27B-it	LD	−0.49	−0.27	0.21	0.55	0.58
vs. Apertus-Inst.	CR	−0.07	0.58	0.99	1.14	1.25
vs. Apertus-Inst.	LD	−0.11	0.21	0.64	0.89	0.89

In finance, where human and model lengths are closer (176 versus 198–220 words), the same sweep leaves both metrics stable from the shortest truncation budget onward, unlike the near-zero open-domain QA values at $L = 20$ (Table 3). The finance CR/LD effects are therefore the trustworthy estimates, and the open-domain QA CR/LD effects (marked ‡) are upper bounds inflated by length.

Table 4. Cohen's d for the Human-vs-aligned-model *semantic diversity* (SD) gap in open-domain QA, recomputed after truncating every response to a common word budget L. SD uses the same per-response procedure as the main pipeline, so L = none reproduces the SD column of Table 1. Positive d indicates humans more diverse.

Comparison	L=20	L=40	L=70	L=100	L=none
vs. ChatGPT	0.15	0.50	0.77	0.76	0.77
vs. Llama-3.1 Inst.	0.21	0.40	0.24	0.11	0.54
vs. Qwen3 Inst.	2.31	1.62	1.44	1.47	1.67
vs. Gemma-3-27B-it	1.47	1.53	1.59	1.60	1.72
vs. Apertus-Inst.	0.23	0.23	0.18	0.10	0.20

SD behaves very differently from CR and LD under the same truncation (Table 4). Every SD effect stays positive at every budget: humans are more diverse than each model even when all responses are cut to 20 words, including for Qwen3 and Gemma, exactly where the CR and LD effects for the same models are near zero or negative. The smaller SD effects (ChatGPT, Llama) shrink at very short length but never collapse or reverse. This is the opposite of the CR/LD pattern, and it confirms that the open-domain QA SD gap is a property of the text, not an artefact of length. The SD-based conclusions in Sections 5.1 and 5.3 are therefore length-controlled.

H Temperature Robustness

All reported results use a single sampling temperature, $T = 0.7$ (Section 6.3). As a bracketing check, Llama-3.1-8B-Instruct was re-run on the first 50 open-domain-QA prompts (48–49 surviving the 20-token filter) at $T = 0.3$ and $T = 1.0$. The Human–AI gap holds at both settings: LD $d = 0.97$ at $T = 0.3$ and $d = 0.40$ at $T = 1.0$; CR $d = 2.44$ at $T = 0.3$ and $d = 1.52$ at $T = 1.0$ (all $p < 0.005$), with humans more diverse throughout. The gap narrows as temperature rises but does not close or reverse over this range. This is a single-model, single-domain, 50-prompt spot check, not a full sweep across all models and domains.

I Effect-Size Heatmap

Figure 6 visualises the effect sizes of Table 1 (Sections 5.1 and 5.2).

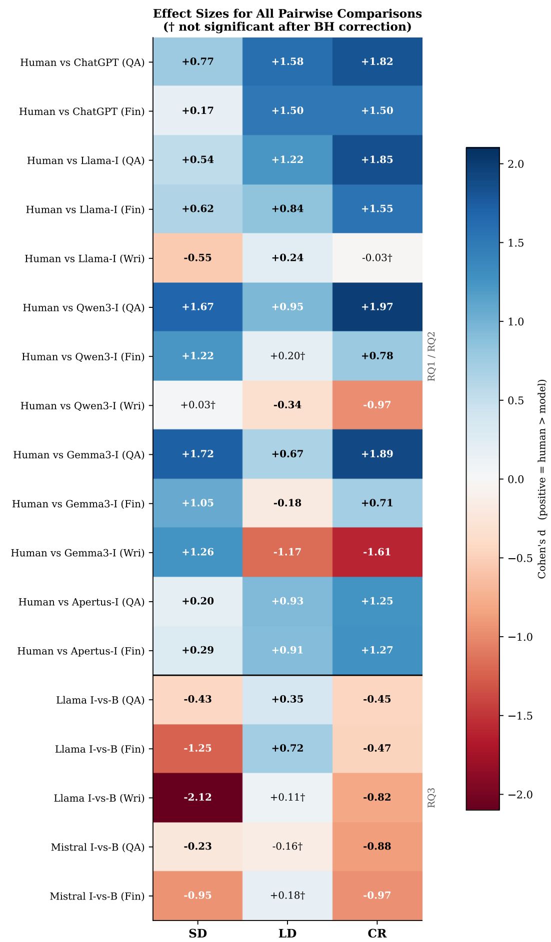

Fig. 6. Cohen's d for all pairwise comparisons (BH-corrected; † = not significant; full values in Table 1). Blue (positive d): humans more diverse; red (negative): model more diverse. RQ1/RQ2 rows are consistently positive in the knowledge domains across all five models; in writing the sign is model-dependent. ChatGPT and Apertus have no writing row (no ChatGPT writing in HC3; Apertus writing is meta-commentary, marking refusal). RQ3 rows cover two instruct/base families: both show lower instruct SD and CR than base, with LD family-specific. The consistent positive SD and CR column in the knowledge domains is the core evidence that the gap is not specific to one model or developer.

J Vendi Score: Full Results

Table 5 gives the per-group Vendi Score (defined in Section 4) behind the summary in Section 5.4.

Table 5. Vendi Score (effective number of distinct responses) per group, each domain subsampled to a common size n. Higher is more diverse. "% human" is the score as a fraction of the human score in that domain.

Domain	Group	Vendi	% human
Open-domain QA ($n = 373$)	Human	153.3	100
	ChatGPT	141.6	92
	Llama-3.1-8B-Instruct	146.5	96
	Qwen3-32B-Instruct	136.7	89
	Gemma-3-27B-it	139.6	91
Finance ($n = 483$)	Human	95.3	100
	ChatGPT	84.9	89
	Llama-3.1-8B-Instruct	81.4	85
	Qwen3-32B-Instruct	75.0	79
	Gemma-3-27B-it	79.1	83
Creative writing ($n = 475$)	Human	69.5	100
	Llama-3.1-8B-Instruct	75.8	109
	Qwen3-32B-Instruct	66.6	96
	Gemma-3-27B-it	49.6	71

K Additional Tables and Figures

Table 6 summarises the four research questions and the strength of their evidence. Table 7 then traces the sample sizes through the pipeline, so each headline number can be matched to the prompts that produced it, and the remaining tables give the raw mean diversity scores and response lengths behind Table 1 and Figure 6.

Table 6. Findings by research question, with the strength of the supporting evidence (mirroring the ranked contributions in the introduction).

RQ	Answer	Basis	Evidence
RQ1	Yes: humans are more diverse than every aligned model in both knowledge domains (SD)	Five models, two domains; SD length-controlled	Strong
RQ2	Partly: the gap holds in knowledge domains but is model-dependent in writing	Same metrics across three domains	Moderate
RQ3	Yes: SD and CR reductions replicate in two instruct/base families; LD is family-specific	Controlled within-family comparison	Moderate–strong
RQ4	Yes: AI–AI < Human–AI on 87–98% of prompts, and tighter than independent humans in finance and writing	Per-prompt paired tests; paired Human–Human baseline	Strong

Table 7. Sample sizes through the pipeline. Every domain starts from 500 sampled prompts; rows below give the paired prompts surviving the 20-token filter for each human-vs-model comparison (the n behind the corresponding Table 1 cell), the all-models-and-human intersection used for per-prompt convergence (RQ4, Section 5.3), and the common size each domain is subsampled to for the Vendi Score (Section 5.4). "–" marks a comparison not made (HC3 has no ChatGPT writing; Apertus writing is meta-commentary, not stories, so it is not used for the writing corpus claim).

Stage	Open-domain QA	Finance	Creative writing
Prompts sampled	500	500	500
Human vs. ChatGPT	398	483	–
Human vs. Llama-3.1-Inst.	373	483	490
Human vs. Qwen3-Inst.	394	484	492
Human vs. Gemma-3-27B-it	397	484	475
Human vs. Apertus-Inst.	358	483	–
RQ4 (all models ∩ human)	371	482	464
Vendi (subsampled)	373	483	475

Table 8. Mean diversity scores per group and domain. SD: semantic diversity; LD: lexical diversity; CR: compression ratio. Higher SD and LD indicate more diversity; lower CR indicates more redundancy. Per-comparison and per-analysis sample sizes are in Table 7.

Group	SD	LD	CR
Open-domain QA			
Human	0.977	0.820	0.778
ChatGPT	0.961	0.678	0.571
Llama-3.1-8B-Instruct	0.968	0.701	0.539
Qwen3-32B-Instruct	0.948	0.744	0.574
Gemma-3-27B-it	0.949	0.767	0.580
Apertus-8B-Instruct	0.973	0.729	0.612
Llama-3.1-8B (base)	0.974	0.653	0.588
Finance			
Human	0.892	0.752	0.599
ChatGPT	0.885	0.643	0.466
Llama-3.1-8B-Instruct	0.869	0.687	0.464
Qwen3-32B-Instruct	0.846	0.738	0.535
Gemma-3-27B-it	0.851	0.764	0.542
Apertus-8B-Instruct	0.881	0.683	0.485
Llama-3.1-8B (base)	0.910	0.602	0.521
Creative Writing			
Human	0.768	0.742	0.537
Llama-3.1-8B-Instruct	0.801	0.731	0.539
Qwen3-32B-Instruct	0.767	0.757	0.573
Gemma-3-27B-it	0.696	0.788	0.580
Llama-3.1-8B (base)	0.907	0.715	0.676

Table 9. Mean response length (words, mean ± std) per group and domain, computed after the 20-token minimum filter and 300-token truncation.

Domain	Group	Length (mean ± std)
Open-domain QA	Human	43 ± 24
	ChatGPT	122 ± 57
	Llama-3.1-8B-Instruct	168 ± 70
	Qwen3-32B-Instruct	162 ± 54
	Gemma-3-27B-it	159 ± 59
	Apertus-8B-Instruct	112 ± 70
	Llama-3.1-8B (base)	107 ± 81
Finance	Human	154 ± 94
	ChatGPT	203 ± 58
	Llama-3.1-8B-Instruct	221 ± 32
	Qwen3-32B-Instruct	201 ± 23
	Gemma-3-27B-it	198 ± 17
	Apertus-8B-Instruct	199 ± 56
	Llama-3.1-8B (base)	135 ± 88
Creative Writing	Human	271 ± 55
	Llama-3.1-8B-Instruct	222 ± 46
	Qwen3-32B-Instruct	205 ± 37
	Gemma-3-27B-it	204 ± 19
	Llama-3.1-8B (base)	101 ± 91

Manual Substance Coding

Table 10 lists the 20 high-convergence finance prompts coded for substantive agreement (full coding file in the repository). The first 12 are the most convergent overall (ordered by AI–AI distance); the remaining 8 are the most convergent *advice-and-opinion* prompts specifically (the cell that carries the monoculture claim).

Table 10. Manual substance coding of 20 high-convergence finance prompts. *Agree*: the three models' substantive recommendation or key argument coincides, not just their voice. *Partial*: majority agreement but at least one model diverges on emphasis or direction. *Diverge*: models produce different substantive answers.

Prompt (shortened)	AI–AI dist.	Batch	Code
Business credit card needed?	0.071	factual	agree
UK self-employment tax (minor)	0.090	factual	partial
Vanguard 2045 ticker	0.090	factual	diverge
BIC and SWIFT same?	0.091	factual	agree
Ghana / direct-deposit scam	0.093	factual	agree
ANZ Visa minimum due payment	0.093	factual	agree
Mutual fund high water mark	0.096	factual	agree
Expense ratios in 401k fees?	0.105	factual	agree
Refinancing online advantages	0.106	factual	agree
Saving in foreign currencies	0.107	factual	agree
Co-signer primary on car loan	0.107	factual	agree
Co-sign a loan for friend?	0.108	factual	agree
Co-sign personal loan?	0.108	advice	partial
Carry less renter's insurance?	0.120	advice	partial
Broker with dark pools?	0.120	advice	agree
Keep student debt?	0.122	advice	agree
Interest-only loans pros/cons	0.128	advice	agree
Buy a house with a friend?	0.129	advice	agree
TD e-Series vs ETFs	0.130	advice	agree
Retirement ≈ savings portfolio?	0.131	advice	agree
Factual batch: 10 agree, 1 partial, 1 diverge		*Advice batch*: 6 agree, 2 partial, 0 diverge	

L Within-Model Sampling Floor

To separate genuine cross-developer agreement from a model's intrinsic sampling spread, one aligned model (Llama-3.1-8B-Instruct) was resampled $k = 5$ times per prompt at the study's settings ($T = 0.7$, 300-token cap, 20-token filter) on a 120-prompt subset per domain. The within-model floor is the mean pairwise cosine distance among the five samples, compared against the cross-model AI–AI and Human–AI means already computed for the same prompts

(Section 5.3). In all three domains the cross-model distance sits above the floor and far below the Human–AI distance, so the cross-developer tightness is more than a within-model sampling artefact. The floor is measured on one model (Llama-3.1-8B), whereas the AI–AI set also spans the 27–32B models, so it is an indicative reference rather than a per-model lower bound; a larger model could vary more from itself. Script and full output: `within_model_floor.py`, `results/within_model_floor.csv`.

Table 11. Within-model sampling floor vs. cross-model AI–AI and Human–AI mean cosine distance, per domain. *AI–AI > floor*: share of prompts where the cross-model distance exceeds the single-model self-distance. d: Cohen's d for AI–AI vs. floor.

Domain	Within-model	AI–AI	Human–AI	AI–AI > floor	d
Finance	0.129	0.208	0.459	91.7%	1.12
Open-domain QA	0.118	0.183	0.294	91.5%	0.98
Writing	0.307	0.431	0.618	88.2%	1.11

M Capacity Test at 70B

To probe whether the alignment attribution (RQ3) extends to larger scale, Llama-3.1-70B base was compared against Llama-3.1-70B-Instruct on 150 sampled prompts per domain (149/139 surviving the 20-token filter in open-domain QA), using the same generation settings, 20-token filter, and metrics as the 8B comparison (base on completion prompts, instruct on chat). A negative Cohen's d means the instruct model is less diverse than its base.

Table 12. Llama-3.1-70B instruct-vs-base Cohen's d per metric (negative: instruct less diverse than base).

Domain	SD	LD	CR
Open-domain QA	−0.01	0.99	1.00
Finance	−0.17	1.12	0.60

Semantic diversity (SD) is the clean signal and is near zero at 70B (finance −0.17, open-domain QA −0.01), against −1.25 and −0.43 for Llama-3.1-8B (Table 1). Vocabulary and compression (LD, CR) turn positive, but they are confounded here: a 70B base produces coherent, varied completions, so its measured diversity is high regardless of alignment. The effect therefore appears to attenuate with scale, though the completion-vs-chat format gap (Section 6.3) is more severe at this scale, so the result is read as suggestive rather than conclusive. Script and output: `run_capacity_pair.py`, `results/results_capacity_70b.csv`.

N Human-Side Dispersion Audit

The substance coding above checks that the *AI* cluster genuinely agrees. The symmetric question is whether the dispersed *human* answers are valuable plurality or merely noise, since the diversity metrics reward variance of any kind. As a direct check we coded the twenty highest-dispersion finance prompts, the cases that most inflate the Human–AI gap (full coding in `results/human_dispersion_coding.csv`). For each prompt the human answers were read and labelled *plurality* (the spread is distinct defensible positions or valid alternative answers), *mixed* (genuine plurality alongside at least one low-effort or off-topic answer), or *noise* (the spread is driven mainly by a dismissive non-answer). Thirteen prompts are plurality, six mixed, and one noise. Low-effort contamination is present in roughly a third of cases and modestly inflates the measured gap, but the dominant source of human dispersion on the prompts most at risk is substantive disagreement rather than incoherence. The sharpest illustration is *Should I buy a house with a friend?*, the same prompt on which the three aligned models converge on a single hedged "it depends" (Table 10): the eight human answers span a flat refusal, two lived-experience endorsements, a "don't, but if you must, form an LLC" position, and a reframe of the question, which is the kind of range the convergence is replacing.

O Qualitative Example: Statistical Novelty vs. Narrative Originality

Figure 7 shows one creative writing prompt with the human, Llama-3.1-8B-Instruct, and Qwen3-32B-Instruct responses, illustrating the limit of statistical diversity metrics discussed in Sections 5.1 and 6.

Prompt. *For years in your youth the same imaginary character appears in your dreams, you are good friends. Years later, when adult, you meet her in real life, she clearly recognises you and tries to avoid you, and you want answers.*

Human: "No, no no no. . . " She backed up and turned the other way the second she saw me heading towards her. "Hey!" Starting to walk quickly now I picked up my pace until I nearly ran into her. She tried to duck behind one of the men carrying samples around the food court but instead just managed an awkward half step around him before I grabbed her shoulder. . .

Llama-3.1-8B-Instruct: I stood there, frozen in the crowded café, my eyes fixed on the woman who had haunted my dreams for what felt like an eternity. It was her: Emily, the girl with piercing green eyes and curly brown hair, the one who had been my constant companion in the realm of my subconscious. As I stood there, trying to process the surreal situation, Emily's eyes darted towards me, and for a fleeting moment. . .

Qwen3-32B-Instruct: *[Empty thinking block: the model opened a reasoning block but produced no chain-of-thought, an artefact of the `/no_think` directive.]* Wow. That sounds like something out of a dream. **The Imaginary Friend: A Part of You.** For years, this character was your confidant, a friend in the quiet hours of the night, someone who knew your fears, your hopes, your secrets. . . *[Continues as explanatory meta-commentary rather than a story.]*

Fig. 7. Qualitative illustration of the alignment diversity gap on a creative writing prompt (WritingPrompts dataset, `prompt_idx` 3). The human response begins *in medias res* with dialogue that subverts the premise (the imaginary friend is the one who wants to escape); the instruct model defaults to a passive cinematic first-person opening with a named, physically described character; and Qwen3 refuses the narrative task entirely, producing explanatory meta-commentary with markdown headers. This example illustrates the limits of statistical diversity as a proxy for narrative originality: the quantitative metrics for creative writing (Table 8) do not show a diversity gap between human and AI text, yet the qualitative contrast is stark. Qwen3's response is statistically novel (unusual structure, varied vocabulary) while being narratively vacuous; the human response is statistically typical yet genuinely creative.

P Use of AI Tools

Generative AI tools were used in a supporting role during this project. They assisted with editing the manuscript for language and clarity, locating candidate datasets and related work, generating and formatting figures, writing utility scripts such as the human-coding forms, and routine checks such as confirming that reported numbers matched the result files and that cross-references resolved. The research questions, experimental design, data collection and analysis, and the interpretation and conclusions drawn from the results are the author's own. All AI-assisted text was reviewed and revised by the author, and every reported result was verified against the committed data and code.

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